

Micro Energy Harvesting 2026

Exercise 5: Capacitors, batteries, efficiencies and leakage currents

Task 1: Radio Messages

You want to supply a radio system from a charged capacitor. For the transmission of a protocol including running the microcontroller you need 200 μJ . The MC is connected via a two-point switch at a capacitor voltage of 2.5 V and disconnected again at 1.9 V. What is the minimum capacity of the capacitor?

$$E = \frac{1}{2} C (U_1^2 - U_2^2)$$

$$C = 2 E / (U_1^2 - U_2^2) = 2 * 200 \mu\text{J} / (6.25 - 3.61) \text{ V}^2 = \underline{151,5 \mu\text{F}}$$

Task 2: NiMH Batteries

A NiMH battery has a Coulomb efficiency η_C of 68% and an energy efficiency η_E of 59%. Its end-of-discharge voltage is 1.0 V and its end-of-charge voltage is 1.5 V. The capacity C_0 of the cell is 15 mAh, the self-discharge rate SDR is 0.2 %/day.

- a) How many hours can you run a low power system continuously on this battery, having the following properties, assuming that no self-discharge occurs during this period:

Operating voltage (kept constant in the system):	1,0 V
Operating power (constant):	150 μW

Calculate the current of the system, then time. The value in mAh is the current measured at a load resistor during discharge, i.e. the actual usable capacity.

$$I = P/U = 150 \mu\text{W} / 1 \text{ V} = 150 \mu\text{A} = 0,150 \text{ mA}$$

$$t = C_0/I = 15 \text{ mAh} / 0,150 \text{ mA} = \underline{100 \text{ h}}$$

- b) What current would an energy harvesting system have to provide to at least compensate for the continuous self-discharge of the battery? Note the corresponding Efficiency.

Note the Coulomb efficiency, by which must be multiplied with the charge current.

The current is dC/dt

$$dC = 0,002 * 15 \text{ mAh} = 30 \mu\text{Ah}$$

$$dt = 24 \text{ h}$$

$$I = dC/dt = 1.25 \mu\text{A}$$

We need $1 / \eta_C = 0,68$ more current to maintain the charge.

$$I = \underline{1,83 \mu\text{A}}$$

- c) What charge quantity Q is required to fully charge the battery once?

$$Q = 15 \text{ mAh} * 1 / 0,68 = \underline{22,5 \text{ mAh}}$$

- d) What current would an energy harvesting system have to provide to keep the battery charged and to supply the low power system continuously. The low power system is supplied directly from the generator. What would change if it was also fed from the battery?

$$I_1 = 1,83 + 150 \mu A = \underline{151,83 \mu A}$$

$$I_2 = 1,83 + 150/0,68 \mu A = \underline{222,43 \mu A}$$

Task 3: Li-Ion Battery

A lithium-ion battery has a Coulomb efficiency η_C of 98 % and an energy efficiency η_E of 90 %. Its end-of-discharge voltage is 3.3 V, its end-of-charge voltage 4.1 V. In this range the battery behaves linear (C-V). The capacity C_0 of the cell is 1.5 Ah, the self-discharge rate SDR is 1.9 %/month (30 d).

- a) Determine the equivalent electrical resistance (EPR) of the battery from the self-discharge rate and the capacity.

This is NOT the internal resistance of the battery, but the theoretical parallel resistance through the cell!

Linear behaviour in the range 4,1 – 3,3 V:

$$\rightarrow C(4,1V) = 1,5 \text{ Ah}; C(3,3V) = 0$$

$$\text{gradient: } dC/dU = 1,5/(4,1-3,3) = (1,5 \text{ Ah})/(0,8 \text{ V}) = 1,875 \text{ Ah/V}$$

Charge loss per month = Current

$$I = dC/dt = (0,019 \cdot 1,5)/(24 \cdot 30) = 39,6 \mu A$$

Voltage loss per month, with fully charged battery (4.1 V) therefore:

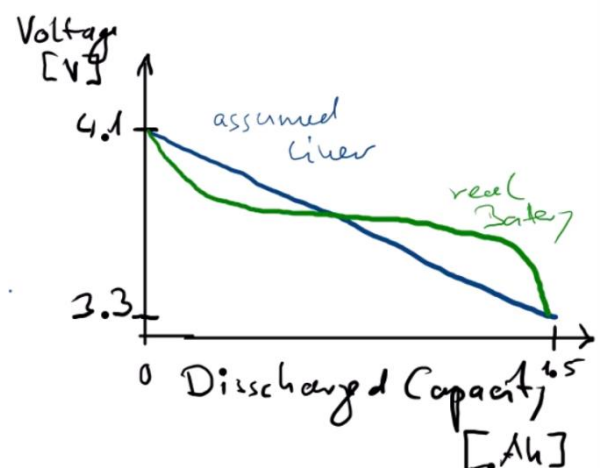
$$dU = (dC/dt)/(dC/dU) = 0,0285/1,875 = 0,0152 \text{ V}$$

Average voltage during discharge: since the profile is linear, we use the mean voltage:

$$U_d = U_{\max} - dU/2 = 4,1 - 0,0152/2 = 4,0924 \text{ V}$$

The equivalent resistance can then be calculated from $U = R \cdot I$

$$EPR = U_d/I = 4,0924V/39,5\mu A = 103386 \Omega$$



- b) What current and power would an energy harvesting generator have to provide to at least compensate for the continuous self-discharge of the battery?

$$I = dC/dt = (0,019 \cdot 1,5) / (24 \cdot 30) = 39,6 \mu A$$

$$I = 39,6 \mu A \cdot 1 / \eta_C = 40,4 \mu A - \text{Taking into account the Coulomb efficiency } \eta_C \text{ of } 98 \%$$

$$P = U \cdot I = 40,4 \cdot 4,1 = \underline{165,7 \mu W}$$

- c) How much energy is required to fully charge the battery once?

U_{Av} during charging

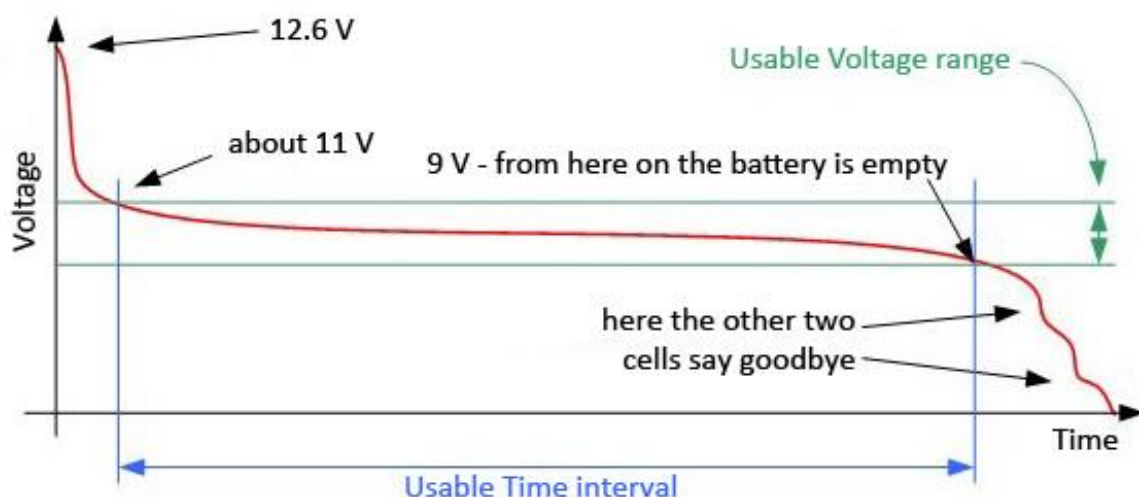
$$U_{av} = (U_{Start} + U_{End}) / 2 = (3.3 \text{ V} + 4.1 \text{ V}) / 2 = 3.7 \text{ V (Linear Characteristic)}$$

$$E = U_{av} \cdot C_0 \cdot 1 / \eta_E = 3,7 \text{ V} \cdot 1.5 \text{ Ah} \cdot 1 / 0.9 = \underline{6,16 \text{ Wh}}$$

Task 4: LiPo-Battery with three cells

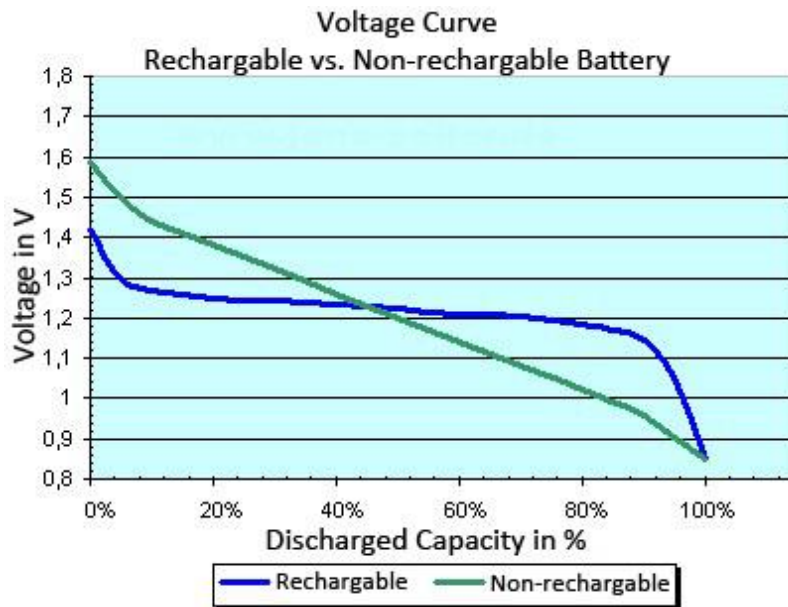
The picture below shows the discharge curve of a three cell LiPo battery. A single cell has a usable voltage between 3.8 and 4.0 V. How would you determine the state of charge of the battery?

3S-LiPo Discharge Curve



Ideally by a microcontroller which has stored the discharge curve in memory. However, this also requires quite exact voltage measurements. Or one measures the amount of charge removed. Temperature measurement, or heat quantity measurement!

Why are Li-Ion batteries and Li-Po better than NiMH batteries? How does it compare to non-rechargeable batteries?



The voltage plateau is advantageous for operating systems. In the case of the battery, the linear drop allows a simple residual capacity measurement. NiMH has a much higher self-discharge and lower capacity.